

DATA AUGMENTATION

Neural Networks

VECTOR DATA AUGMENTATION



$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} \rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} N(0, \sigma_1^2) \\ N(0, \sigma_2^2) \\ N(0, \sigma_3^2) \\ \vdots \\ N(0, \sigma_n^2) \end{bmatrix}$$

IMAGE DATA AUGMENTATION

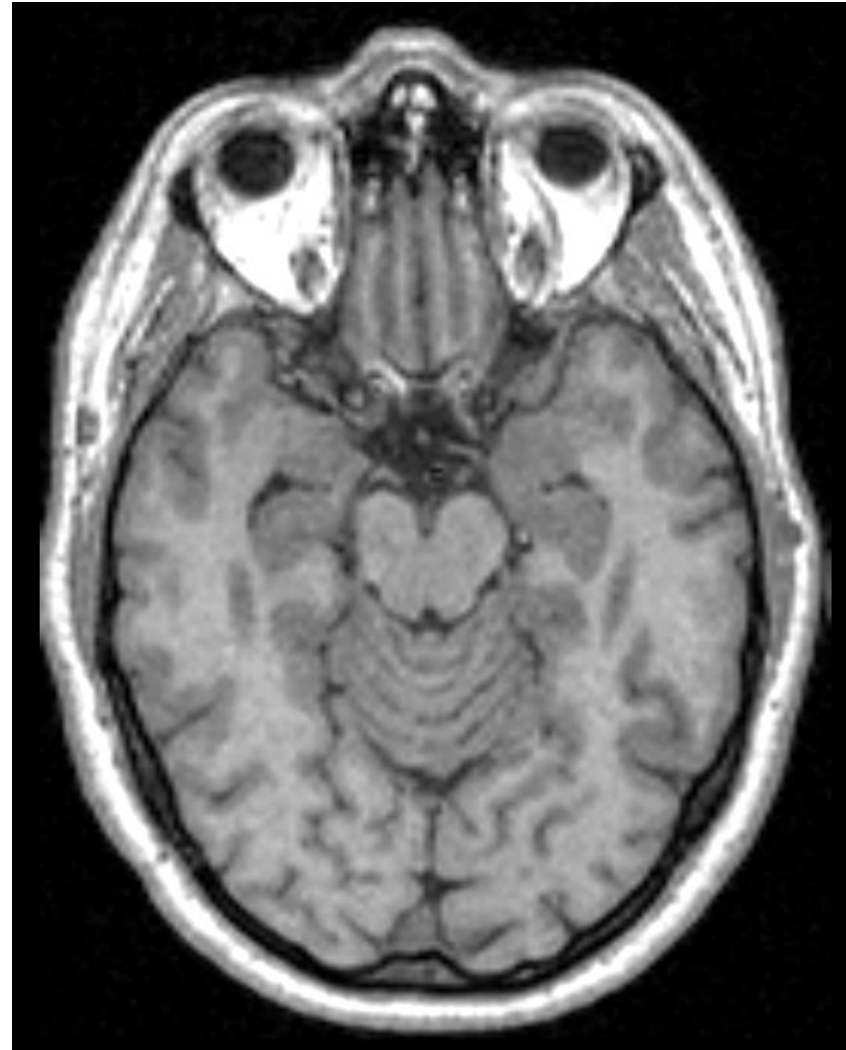
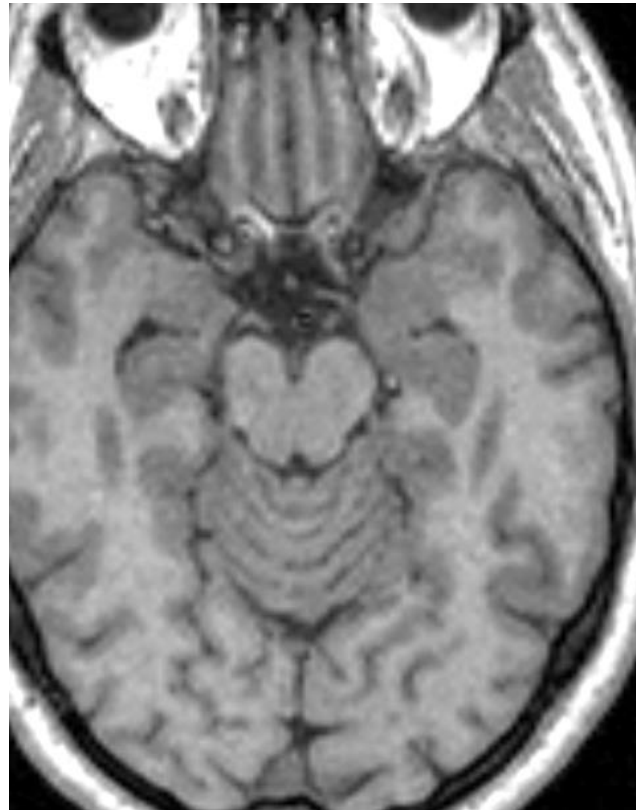


IMAGE DATA AUGMENTATION



Random noise

IMAGE DATA AUGMENTATION



Random crop

IMAGE DATA AUGMENTATION

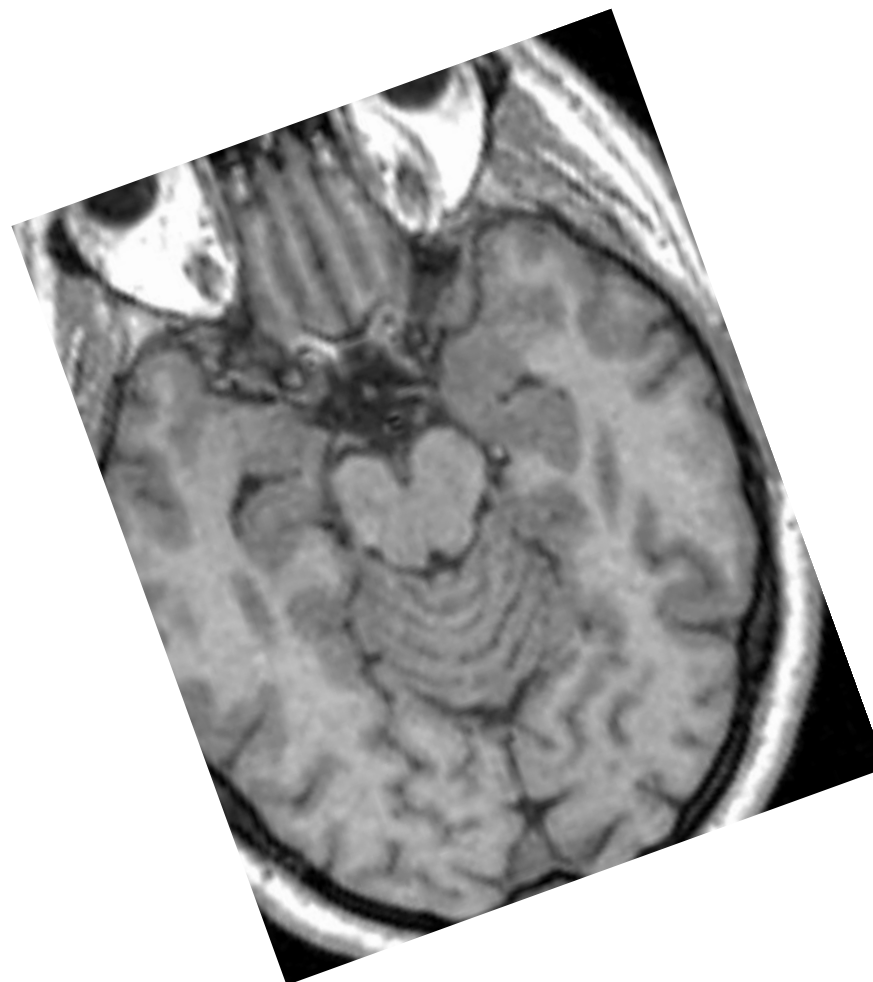


Random zoom

IMAGE DATA AUGMENTATION

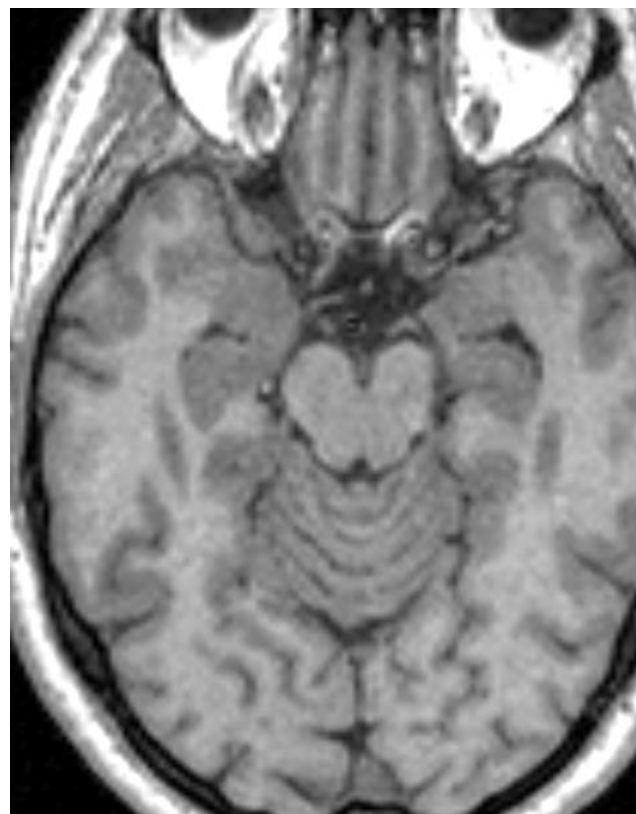


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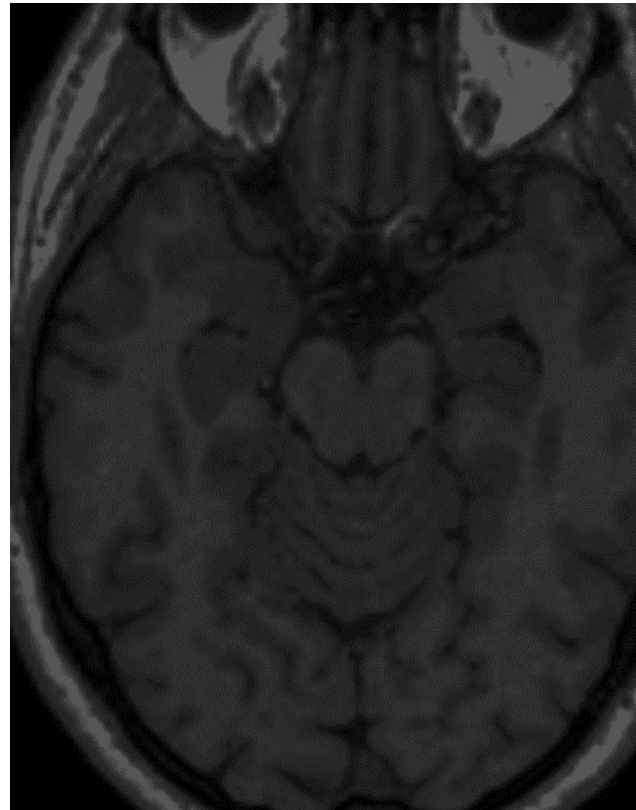
Random rotation

IMAGE DATA AUGMENTATION



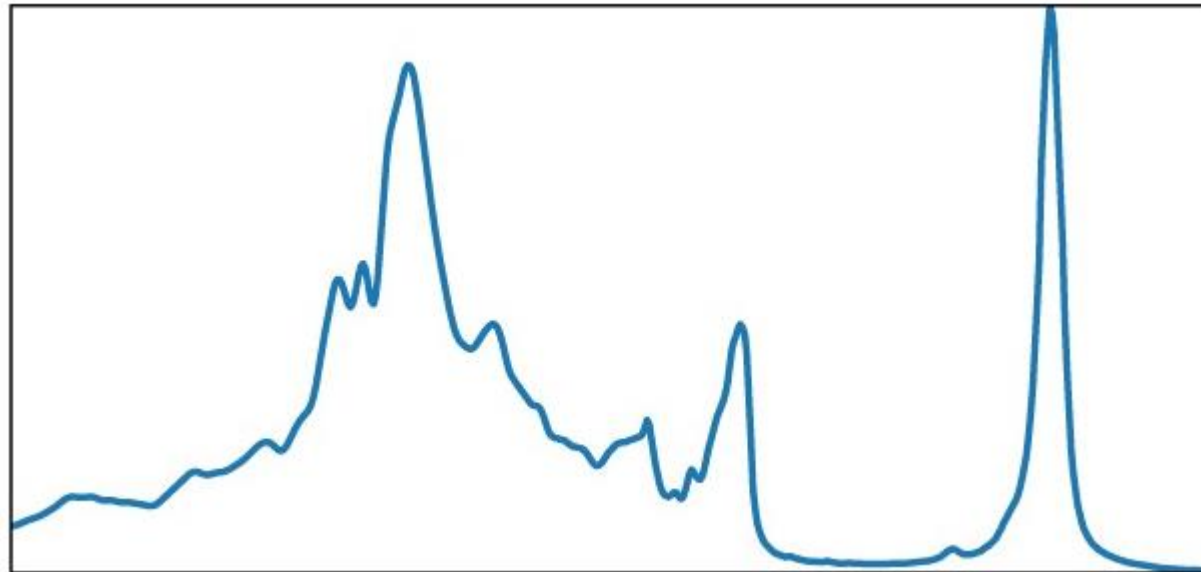
Random flip

IMAGE DATA AUGMENTATION



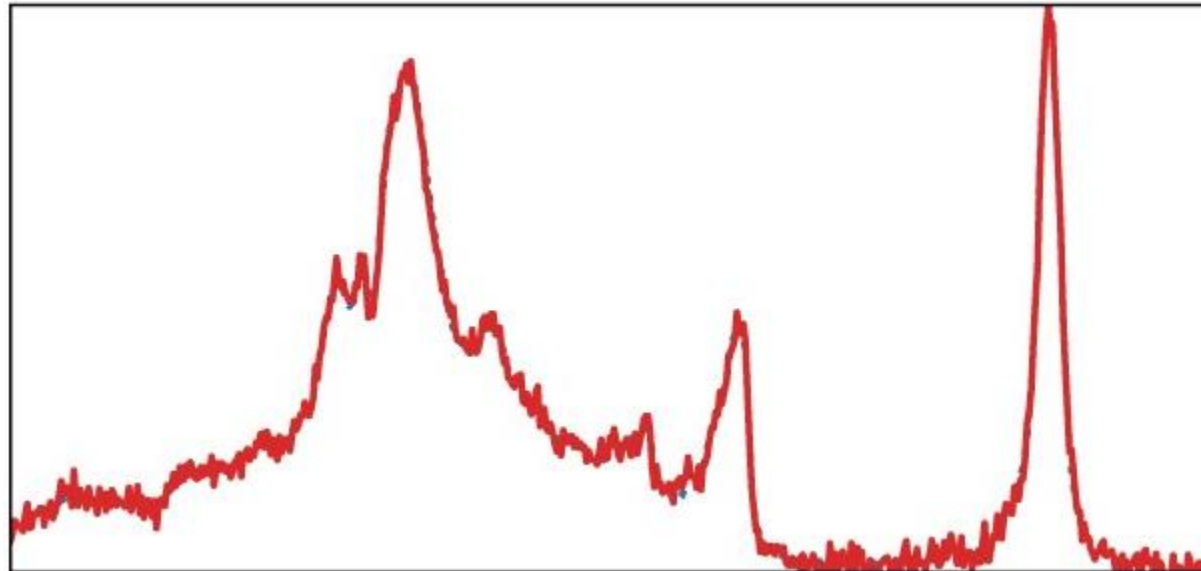
Random brightness

SEQUENCE DATA AUGMENTATION



Iwana & Uchida, PLOS One, 2021.

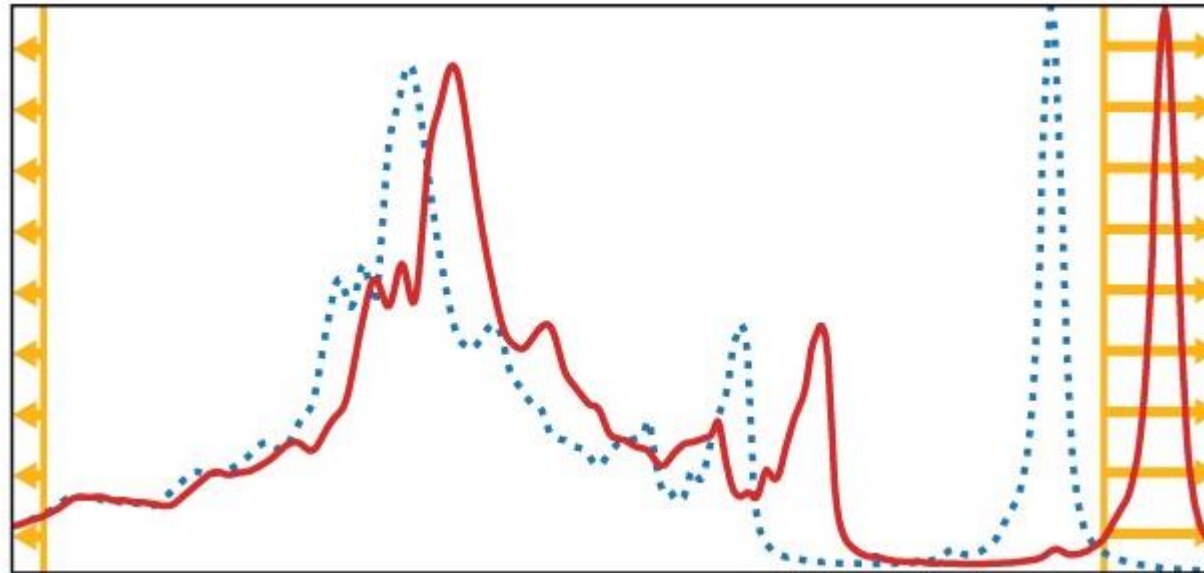
SEQUENCE DATA AUGMENTATION



Random noise

Iwana & Uchida, PLOS One, 2021.

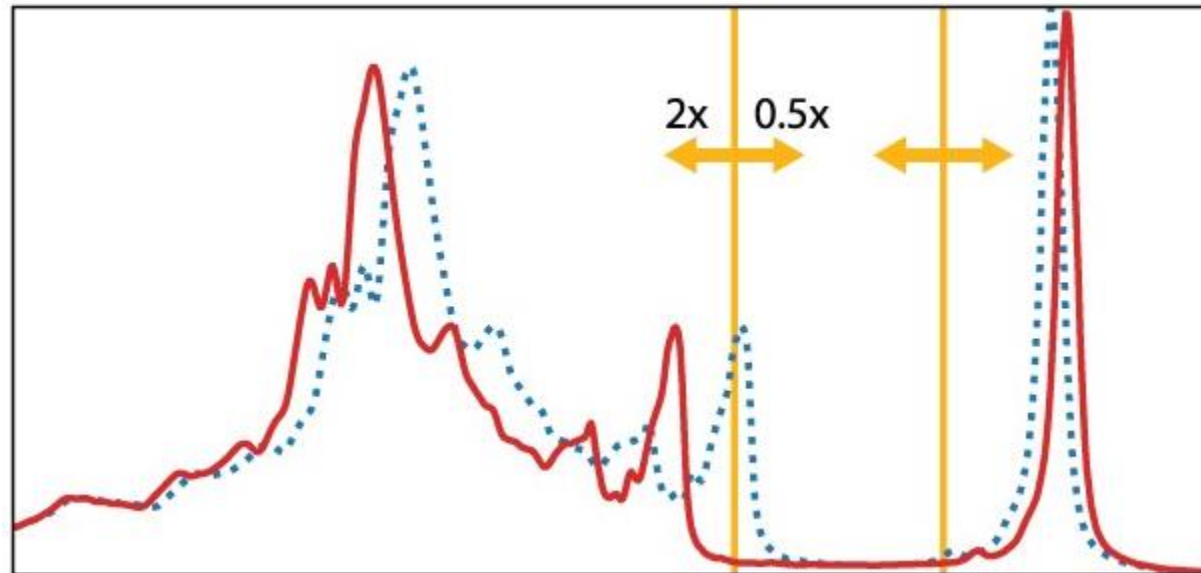
SEQUENCE DATA AUGMENTATION



Window Slicing

Iwana & Uchida, PLOS One, 2021.

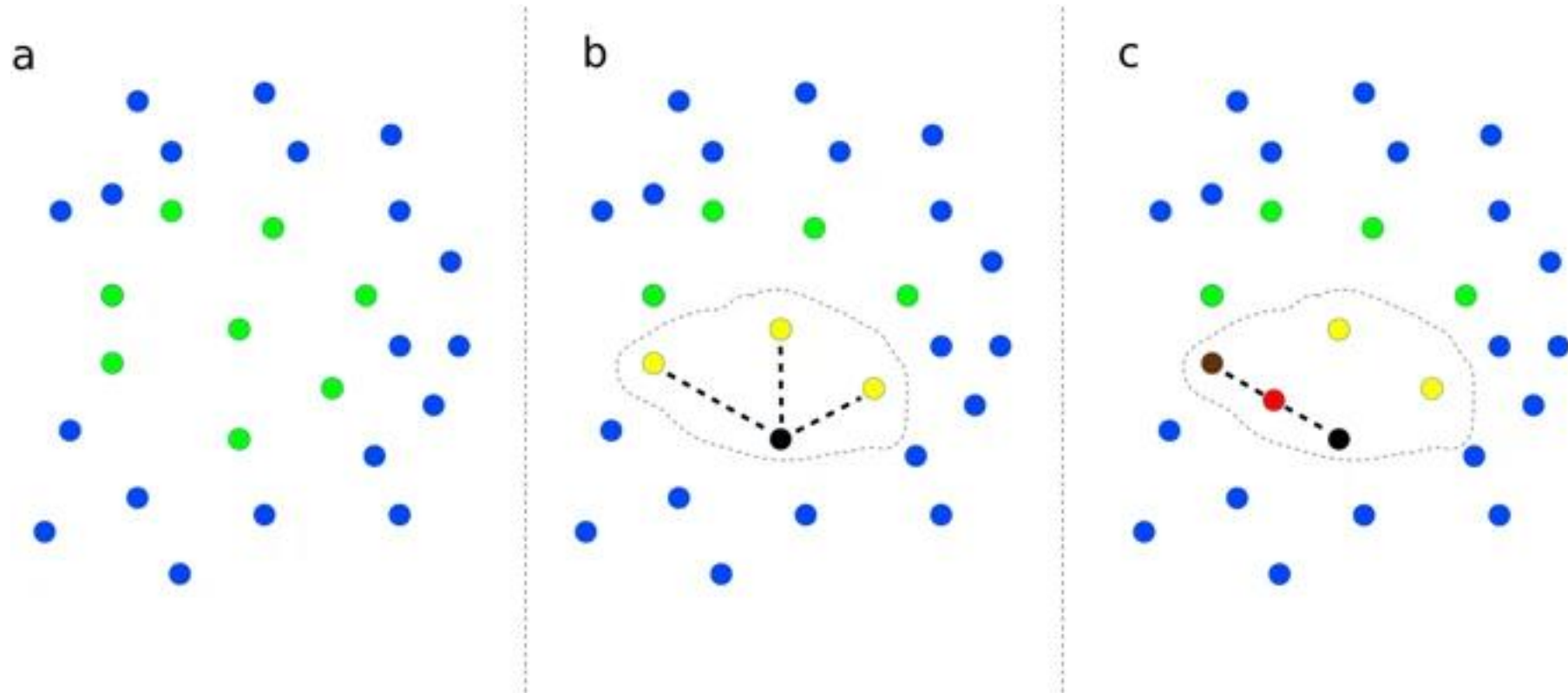
SEQUENCE DATA AUGMENTATION



Window Warping

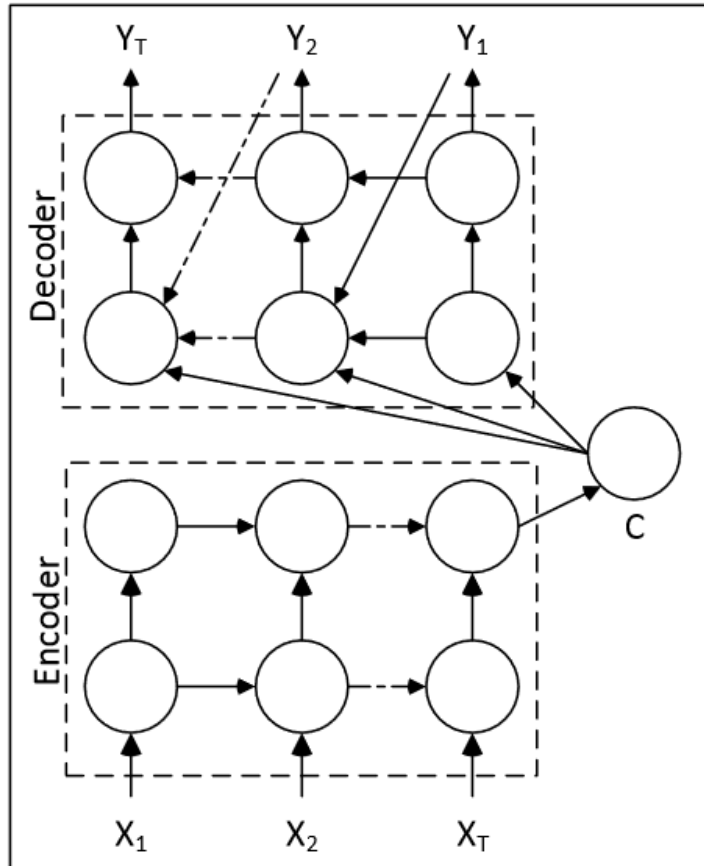
Iwana & Uchida, PLOS One, 2021.

SMOTE

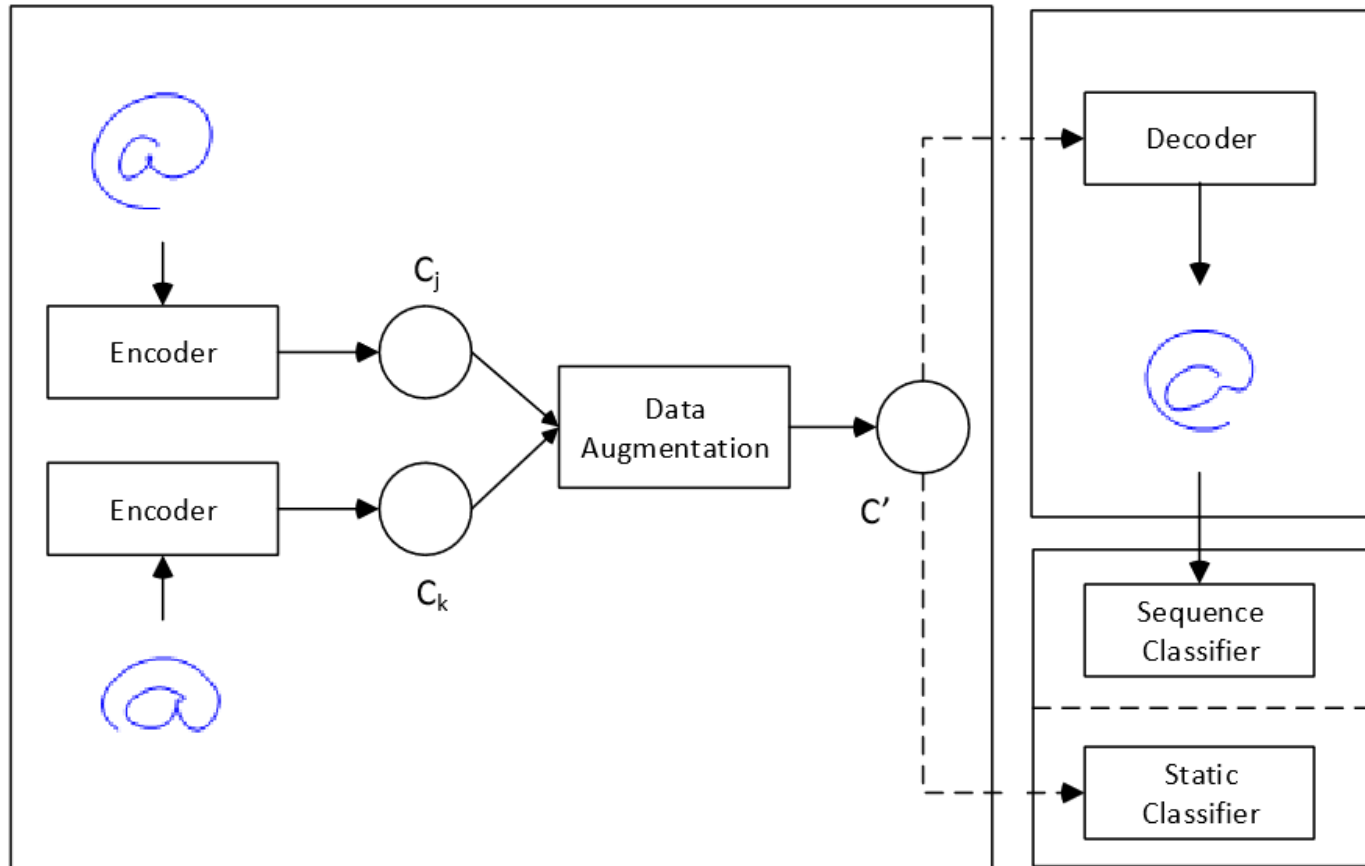


Schubach et al., Nat Sci Rep, 2017.

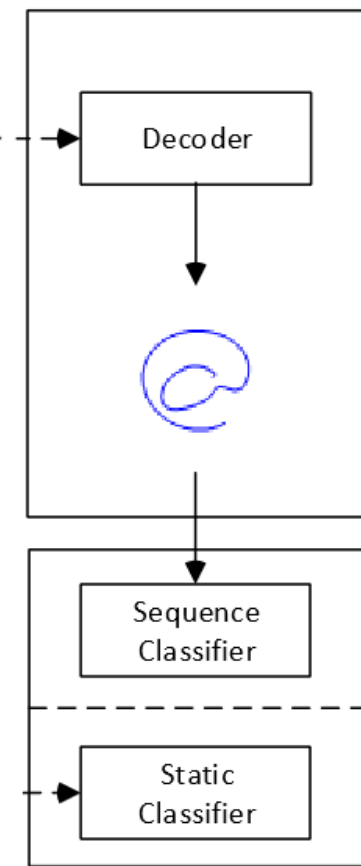
VARIATIONAL AUTOENCODERS



(a) Sequence autoencoder



(b) Encode and apply data transform

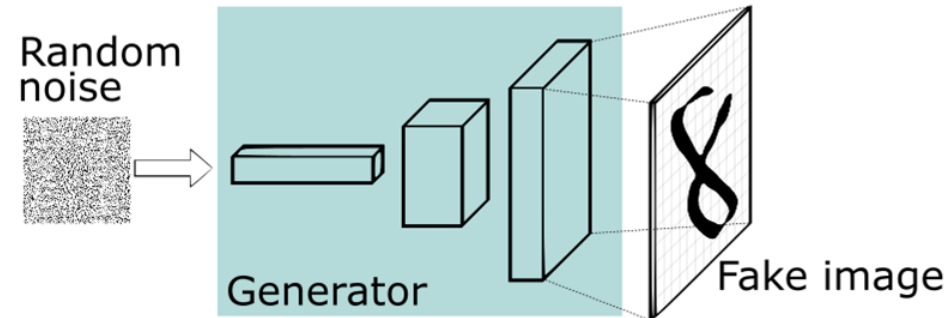


(c) Decode and/or classify

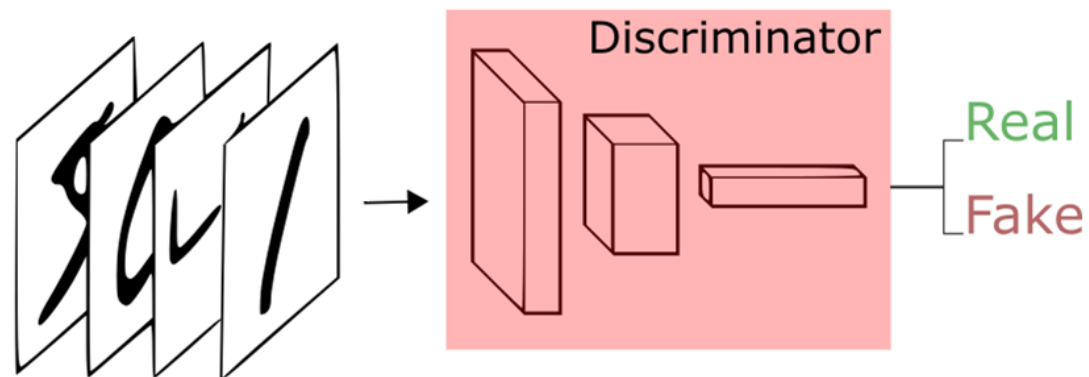
DeVries & Taylor, ICLR Workshop, 2017.

GANs

- Generator

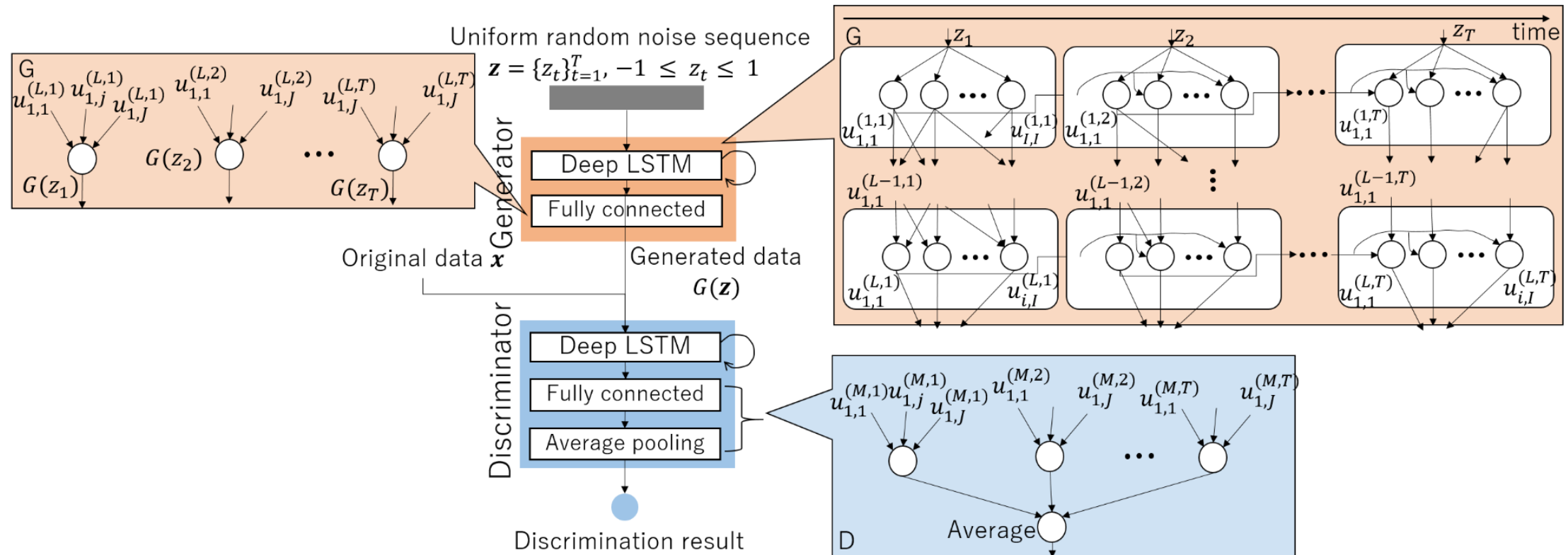


- Discriminator



Source: <https://pathmind.com/wiki/generative-adversarial-network-gan>

GANS



Harada et al., EMBC, 2018.